Konstantinos Kanellopoulos

konkanello@gmail.com konkanello Google Scholar DBLP konkanello.github.io

Education

2020–2026 **Ph.D. Candidate**, *SAFARI Research Group, ETH Zurich*
Advisor: Prof. Onur Mutlu
Thesis: *"Cutting Through the Virtual Memory Wall: From Rigid Abstractions to High-Performance, Flexible, Efficient Cooperation Between Hardware and System Software"* [slides]

2018–2020 **Research Assistant**, *SAFARI Research Group, ETH Zurich*
Advisor: Prof. Onur Mutlu

2012–2017 **Diploma in Computer Science** (5-year joint degree, 300 ECTS), *School of Electrical and Computer Engineering, NTUA*
Major: Computer Systems & Software **Minor:** Computer Networks
Advisor: Prof. Georgios Goumas
Thesis: *"Optimization Techniques for Applications with Irregular Memory Accesses"*

2006–2012 **High School Diploma**, *Greek-French School of Ursulines*

Professional Experience

Summer 2025 **Research Intern**, *Microsoft Research*, Cambridge, UK
Hardware/OS co-design for efficient scheduling of Physical AI applications
Mentors: Bozidar Radunovic, Xenofon Foukas, Ganesh Ananthanarayanan

2020–2021 **Visiting Researcher**, *Huawei Research*, Zurich, Switzerland
Optimizing the address translation subsystem
Mentor: Davide Basilio Bartolini

Honors and Recognitions

Distinguished Artifact Award, HPCA **2026**
Athena: Synergizing Data Prefetching and Off-Chip Prediction via Online Reinforcement Learning [news]

SIGMICRO Hall of Fame **2024**
Included after 8 publications at MICRO [news]

Best Paper Award, ISCA **2024**
Constable: Improving Performance and Power Efficiency by Safely Eliminating Load Instruction Execution [news]

Qualcomm Innovation Fellowship Europe, Finalist **2024**
18 of 62 proposals selected for the final round

Best Paper Award, HiPEAC **2023**
MetaSys: A Practical Open-source Metadata Management System to Implement and Evaluate Cross-layer Optimizations [news]

Distinguished Artifact Award, MICRO **2023**
Victima: Drastically Increasing Address Translation Reach by Leveraging Underutilized Cache Resources [news]

3rd Place, Student Research Competition, MICRO **2023**
Graduate Category

Best Paper Award, MICRO **2022**
Hermes: Accelerating Long-Latency Load Requests via Perceptron-Based Off-Chip Load Prediction [news]

Intel Hardware Security Academic Award, Finalist **2022**
BlockHammer: Preventing RowHammer at Low Cost by Blacklisting Rapidly-Accessed DRAM Rows [news]

Floyd-Warshall Parallel Programming Challenge **2016**
Best execution time, largest input (Parallel Systems, NTUA)

"The Great Moment for Education" Eurobank Scholarship **2012**
Best performance in National qualification exams

Mathematics Competition, Greek-French Schools **2009**
3rd place among 150 students

Peer-Reviewed Publications

Please visit DBLP and Google Scholar for a complete list of publications.

1. K. Kanellopoulos, K. Sgouras, H. Songara, R. Bera, O. Mutlu
"Trail: Effective, Low-Cost and Scalable Temporal TLB Prefetching via Page-Table-Embedded Deltas"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Athens, Greece, 2026
MICRO 2026 [code] [webpage]
Officially artifact evaluated as available, functional, and reproduced.

2. H. Songara, K. Kanellopoulos, F. N. Bostanci, K. Sgouras, A. Olgun, I. E. Yuksel, A. K. Kakolyris, A. G. Yaglikci, O. Mutlu
"Terracotta: Enabling DRAM Techniques via a Flexible DRAM Interface and Memory Controller"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Athens, Greece, 2026
MICRO 2026
Officially artifact evaluated as available, functional, and reproduced.

3. V.-P. Nitu, H. Songara, K. Sgouras, S. Galanopoulos, K. Kanellopoulos, O. Mutlu
"Argus: Agentic, Reference-Calibrated, Tree-Guided, System-Software-Level Bottleneck Localization"
In the Architecture 2.0 Workshop at ISCA, Raleigh, North Carolina, USA, 2026
Architecture 2.0 Workshop, ISCA 2026

4. A. K. Kakolyris, F. N. Bostanci, A. Olgun, I. E. Yuksel, K. M. Sgouras, H. Songara, U. Baser, K. Kanellopoulos, A. G. Yaglikci, O. Mutlu
"ColumnKeeper: Efficient Solutions for Mitigating ColumnDisturb in DRAM-based Systems"
In Proceedings of the ACM/IEEE International Symposium on Computer Architecture (ISCA), Raleigh, North Carolina, USA, 2026
ISCA 2026
Officially artifact evaluated as available, functional, and reproduced.

5. K. Kanellopoulos, K. Sgouras, A. K. Kakolyris, V.-P. Nitu, B. K. Konar, R. Bera, O. Mutlu
"Revelator: Rapid Data Fetching via OS-Driven Hash-based Speculative Address Translation"
In Proceedings of the ACM/IEEE International Symposium on Computer Architecture (ISCA), Raleigh, North Carolina, USA, 2026
ISCA 2026 [pdf] [slides] [code] [webpage]

6. R. Bera, Z. Lang, C. Hengartner, K. Kanellopoulos, R. Kumar, M. Sadrosadati, O. Mutlu
"Athena: Synergizing Data Prefetching and Off-Chip Prediction via Online Reinforcement Learning"
In Proceedings of the IEEE International Symposium on High-Performance Computer Architecture (HPCA), Sydney, Australia, 2026
HPCA 2026 [pdf] [code] [slides]
Officially artifact evaluated as available, functional, and reproduced.
Distinguished Artifact Award at HPCA 2026

7. N. M. Ghiasi, T. Güloglu, H. Mustafa, C. Firtina, K. Koliogeorgi, K. Kanellopoulos, H. Mao, R. Nadig, M. Sadrosadati, J. Park, O. Mutlu
"SAGe: A Lightweight Algorithm-Architecture Co-Design for Mitigating the Data Preparation Bottleneck in Large-Scale Genome Sequence Analysis"
In Proceedings of the IEEE International Symposium on High-Performance Computer Architecture (HPCA), Sydney, Australia, 2026
HPCA 2026 [pdf] [slides]

8. N. Mansouri-Ghiasi, M. Sadrosadati, G. F. Oliveira, K. Kanellopoulos, R. Ausavarungnirun, J. Gómez-Luna, A. Manglik, J. D. Ferreira, J. S. Kim, C. Giannoula, N. Vijaykumar, J. Park, O. Mutlu
"RevaMp3D: Architecting the Processor Core and Cache Hierarchy for Systems with Monolithically-Integrated Logic and Memory"
In ACM Transactions on Architecture and Code Optimization (TACO), 2026
ACM TACO 2026 [pdf]

9. F. N. Bostanci, O. Canpolat, A. Olgun, İ. E. Yüksel, K. Kanellopoulos, M. Sadrosadati, A. G. Yağlıkçı, O. Mutlu
"Understanding and Mitigating Covert Channel and Side Channel Vulnerabilities Introduced by RowHammer Defenses"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Seoul, South Korea, 2025
MICRO 2025 [pdf] [code] [slides]
Officially artifact evaluated as available, functional, and reproduced.

10. F. N. Bostanci*, K. Kanellopoulos*, A. Olgun, A. G. Yaglikci, I. E. Yuksel, N. M. Ghiasi, Z. Bingol, M. Sadrosadati, O. Mutlu
"Revisiting Main Memory-Based Covert and Side Channel Attacks in the Context of Processing-in-Memory"
In Proceedings of the IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), Naples, Italy, 2025
DSN 2025 [pdf] [code] [slides] **Equal contribution*
Officially artifact evaluated as available, reviewed, and reproduced.

11. K. Kanellopoulos, K. Sgouras, N. Bostanci, A. Kakolyris, B. Konar, R. Bera, R. Kumar, M. Sadrosadati, N. Vijaykumar, O. Mutlu
"Virtuoso: Enabling Fast and Accurate Virtual Memory Research via an Imitation-based OS Simulation Methodology"
In Proceedings of the ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Rotterdam, The Netherlands, 2025
ASPLOS 2025 [pdf] [code] [slides] [webpage]

12. F. N. Bostanci, İ. E. Yüksel, A. Olgun, K. Kanellopoulos, Y. C. Tuğrul, A. G. Yağlıkçı, M. Sadrosadati, O. Mutlu
"CoMeT: Count-Min-Sketch-based Row Tracking to Mitigate RowHammer at Low Cost"
In Proceedings of the IEEE International Symposium on High-Performance Computer Architecture (HPCA), Edinburgh, Scotland, 2024
HPCA 2024 [pdf] [code] [slides]
Officially artifact evaluated as available, reviewed and reproducible.

13. O. Canpolat, A. G. Yağlıkçı, A. Olgun, İ. E. Yüksel, Y. C. Tuğrul, K. Kanellopoulos, O. Ergin, O. Mutlu
"BreakHammer: Enhancing RowHammer Mitigations by Carefully Throttling Suspect Threads"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Austin, Texas, USA, 2024
MICRO 2024 [pdf] [code] [slides]
Officially artifact evaluated as available, functional, reusable and reproducible.

14. R. Bera, A. Ranganathan, J. Rakshit, S. Mahto, A. V. Nori, J. Gaur, A. Olgun, K. Kanellopoulos, M. Sadrosadati, S. Subramoney, O. Mutlu
"Constable: Improving Performance and Power Efficiency by Safely Eliminating Load Instruction Execution"
In Proceedings of the ACM/IEEE International Symposium on Computer Architecture (ISCA), Buenos Aires, Argentina, 2024
ISCA 2024 [pdf] [code] [slides]
Best Paper Award at ISCA 2024

15. D. Mishra, K. Kanellopoulos, A. Panwar, A. Sriraman, V. Seshadri, O. Mutlu, T. C. Mowry
"Address Scaling: Architectural Support for Fine-Grained Thread-Safe Metadata Management"
In IEEE Computer Architecture Letters (CAL), 2024
IEEE CAL 2024 [pdf]

16. L. Orosa, S. Koppula, Y. Umuroglu, K. Kanellopoulos, J. Gómez-Luna, M. Blott, K. A. Vissers, O. Mutlu
"EcoFlow: Efficient Convolutional Dataflows for Low-Power Neural Network Accelerators"
In IEEE Transactions on Computers (TC), 2024
IEEE TC 2024

17. K. Kanellopoulos, R. Bera, K. Stojiljkovic, N. Bostanci, C. Firtina, R. Ausavarungnirun, R. Kumar, N. Hajinazar, M. Sadrosadati, N. Vijaykumar, O. Mutlu
"Utopia: Efficient Address Translation using Hybrid Virtual-to-Physical Address Mapping"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Toronto, Canada, 2023
MICRO 2023 [pdf] [code] [slides]

18. K. Kanellopoulos, H. C. Nam, F. N. Bostanci, R. Bera, M. Sadrosadati, R. Kumar, D. B. Bartolini, O. Mutlu
"Victima: Drastically Increasing Address Translation Reach by Leveraging Underutilized Cache Resources"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Toronto, Canada, 2023
MICRO 2023 [pdf] [code] [slides]
Officially artifact evaluated as available, functional, reusable and reproducible.
Distinguished Artifact Award at MICRO 2023

19. K. Kanellopoulos, K. Sgouras, O. Mutlu
"Virtuoso: An Open-Source, Comprehensive and Modular Simulation Framework for Virtual Memory Research"
In the ACM Student Research Competition at MICRO, Toronto, Canada, 2023
MICRO SRC 2023
3rd Place, Graduate Category

20. A. Olgun, J. Gomez-Luna, K. Kanellopoulos, B. Salami, H. Hassan, O. Ergin, O. Mutlu
"PiDRAM: A Holistic End-to-end FPGA-based Framework for Processing-in-DRAM"
In ACM Transactions on Architecture and Code Optimization (TACO), 2023
TACO 2023 [pdf] [code] [slides]

21. H. Gupta, M. Kabra, J. Gómez-Luna, K. Kanellopoulos, O. Mutlu
"Evaluating Homomorphic Operations on a Real-World Processing-In-Memory System"
In Proceedings of the IEEE International Symposium on Workload Characterization (IISWC), Ghent, Belgium, 2023
IISWC 2023

22. N. Mansouri-Ghiasi, N. Vijaykumar, G. F. Oliveira, L. Orosa, I. Fernandez, M. Sadrosadati, K. Kanellopoulos, N. Hajinazar, J. Gómez-Luna, O. Mutlu
"ALP: Alleviating CPU-Memory Data Movement Overheads in Memory-Centric Systems"
In IEEE Transactions on Emerging Topics in Computing (TETC), 2023
IEEE TETC 2023

23. R. Bera, K. Kanellopoulos, S. Balachandran, D. Novo, A. Olgun, M. Sadrosadati, O. Mutlu
"Hermes: Accelerating Long-Latency Load Requests via Perceptron-Based Off-Chip Load Prediction"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Chicago, Illinois, USA, 2022
MICRO 2022 [pdf] [code] [slides]
Officially artifact evaluated as available, functional, and reproduced.
Best Paper Award at MICRO 2022

24. D. S. Cali, K. Kanellopoulos, J. Lindegger, Z. Bingöl, G. S. Kalsi, Z. Zuo, C. Firtina, M. B. Cavlak, J. S. Kim, N. Mansouri-Ghiasi, G. Singh, J. Gómez-Luna, N. A. Alserr, M. Alser, S. Subramoney, C. Alkan, S. Ghose, O. Mutlu
"SeGraM: A Universal Hardware Accelerator for Genomic Sequence-to-Graph and Sequence-to-Sequence Mapping"
In Proceedings of the ACM/IEEE International Symposium on Computer Architecture (ISCA), New York, New York, USA, 2022
ISCA 2022 [pdf] [code] [slides]

25. N. Vijaykumar, A. Olgun, K. Kanellopoulos, F. N. Bostanci, H. Hassan, M. Lotfi, P. B. Gibbons, O. Mutlu
"MetaSys: A Practical Open-source Metadata Management System to Implement and Evaluate Cross-layer Optimizations"
In ACM Transactions on Architecture and Code Optimization (TACO), 2022
TACO 2022 [pdf] [code] [slides]
Best Paper Award at HiPEAC 2023

26. M. Besta, R. Kanakagiri, G. Kwasniewski, R. Ausavarungnirun, J. Beránek, K. Kanellopoulos, K. Janda, Z. Vonarburg-Shmaria, L. Gianinazzi, I. Stefan, J. Gómez-Luna, J. Golinowski, M. Copik, L. Kapp-Schwoerer, S. Di Girolamo, N. Blach, M. Konieczny, O. Mutlu, T. Hoefler
"SISA: Set-Centric Instruction Set Architecture for Graph Mining on Processing-in-Memory Systems"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Virtual, 2021
MICRO 2021 [pdf] [slides]

27. R. Bera, K. Kanellopoulos, A. Nori, T. Shahroodi, S. Subramoney, O. Mutlu
"Pythia: A Customizable Hardware Prefetching Framework Using Online Reinforcement Learning"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Virtual, 2021
MICRO 2021 [pdf] [code] [slides]
Officially artifact evaluated as available, functional, and reproduced.

28. A. G. Yaglikçi, M. Patel, J. S. Kim, R. Azizi, A. Olgun, L. Orosa, H. Hassan, J. Park, K. Kanellopoulos, T. Shahroodi, S. Ghose, O. Mutlu
"BlockHammer: Preventing RowHammer at Low Cost by Blacklisting Rapidly-Accessed DRAM Rows"
In Proceedings of the IEEE International Symposium on High-Performance Computer Architecture (HPCA), Virtual, 2021
HPCA 2021 [pdf] [code] [slides]
Intel Hardware Security Academic Award Finalist 2022

29. N. Hajinazar, P. Patel, M. Patel, K. Kanellopoulos, S. Ghose, R. Ausavarungnirun, G. F. Oliveira, J. Appavoo, V. Seshadri, O. Mutlu
"The Virtual Block Interface: A Flexible Alternative to the Conventional Virtual Memory Framework"
In Proceedings of the ACM/IEEE International Symposium on Computer Architecture (ISCA), Virtual, 2020
ISCA 2020 [pdf] [slides]

30. K. Kanellopoulos, N. Vijaykumar, C. Giannoula, R. Azizi, S. Koppula, N. Mansouri-Ghiasi, T. Shahroodi, J. Gómez-Luna, O. Mutlu
"SMASH: Co-designing Software Compression and Hardware-Accelerated Indexing for Efficient Sparse Matrix Operations"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Columbus, Ohio, USA, 2019
MICRO 2019 [pdf] [code] [slides]

31. S. Koppula, L. Orosa, A. G. Yaglikçi, R. Azizi, T. Shahroodi, K. Kanellopoulos, O. Mutlu
"EDEN: Enabling Energy-Efficient, High-Performance Neural Network Inference Using Approximate DRAM"
In Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO), Columbus, Ohio, USA, 2019
MICRO 2019 [pdf] [slides]

32. K. Kanellopoulos
"Prefetch-Process-Thread-Alternation to Speed Up Dijkstra's Algorithm"
In the MultiProg Workshop at HiPEAC, Manchester, UK, 2018
MultiProg Workshop, HiPEAC 2018

Manuscripts Under Submission

33. K. Kanellopoulos, S. Galanopoulos, K. Sgouras, V. Nitu, I. Papalampou, A. Kakolyris, R. Bera, K. Koliogeorgi, D. Masouros, D. Soudris, O. Mutlu
"Valinor: Architectural Support for Fast, Energy-Efficient and Programmable Physical Memory Allocation"
Under Submission [arXiv]

34. E. H. Demirli, H. Songara, U. Başer, O. Ecemiş, R. Bera, O. Ergin, K. Kanellopoulos, O. Mutlu
"SyLo: Exposing Instruction-Level Parallelism by Symbolically Compacting Loop Representation Streams"
Under Submission

35. K. Sgouras, I. E. Yuksel, H. Songara, F. N. Bostanci, K. Koliogeorgi, N. M. Ghiasi, A. Olgun, K. Kanellopoulos, O. Mutlu
"PolyPUM: High Performance Processing-Using-DRAM via Dynamic Number Representation Selection"
Under Submission

36. Z. Lang, R. Bera, K. Kanellopoulos, O. Mutlu
"LIMO: A Lightweight, Multi-Objective Framework for Jointly Optimizing Microarchitectural Policies via Online Stochastic Gradient Estimation"
Under Submission

Patents / Invention Disclosures

X. Foukas, K. Kanellopoulos, B. Radunovic, G. Ananthanarayanan, L. Cox
"Real-time Telemetry-driven CPU/GPU Collaborative Scheduling for Physical AI"
Microsoft Invention Disclosure, 2025 (patent filing in progress)

Invited Talks

Cutting Through the Virtual Memory Wall: From Rigid Abstractions to High-Performance, Flexible, Efficient Cooperation Between Hardware and System Software
New York University (NYU), New York, US — **July 2026**
Massachusetts Institute of Technology (MIT), Cambridge, US — **July 2026**
ACCESS Seminar, AI Chip Center for Emerging Smart Systems (Remote) — **June 2026**

Rethinking the Hardware-OS-Software Interface Towards Efficient Memory Management and Fast Sparse Matrix Operations
Microsoft Research, Cambridge, UK — **Aug. 2025**

Rethinking the OS/Hardware Interface Towards Efficient Virtual Memory
AMD Research, Austin, US — **Oct. 2024**
The University of Texas at Austin, Austin, US — **Oct. 2024**
Apple, Austin, US — **Oct. 2024**
Microlab, NTUA, Athens, Greece — **June 2024**

Victima: Address Translation via Underutilized Cache Resources
Intel Research India (Remote) — **Oct. 2024**

Efficient Address Translation by Leveraging Underutilized Cache Resources and Hash-based Mappings
University of Toronto, Toronto, Canada — **Sep. 2023**
Huawei Research, Toronto, Canada — **Sep. 2023**

MetaSys: A Practical Open-source Metadata Management System to Implement and Evaluate Cross-layer Optimizations
6th Computing Systems Research Day, NTUA, Athens, Greece — **Jan. 2023**

Paper Presentations

Trail: Effective, Low-Cost and Scalable Temporal TLB Prefetching via Page-Table-Embedded Deltas — **MICRO 2026**

Revelator: Rapid Data Fetching via OS-Driven Hash-based Speculative Address Translation [slides] — **ISCA 2026**

Virtuoso: Enabling Fast and Accurate Virtual Memory Research via an Imitation-based OS Simulation Methodology [slides] — ASPLOS 2025

Utopia: Efficient Address Translation using Hybrid Virtual-to-Physical Address Mapping [slides] — MICRO 2023

Victima: Drastically Increasing Address Translation Reach by Leveraging Underutilized Cache Resources [slides] — MICRO 2023

Virtuoso: An Open-Source, Comprehensive and Modular Simulation Framework for Virtual Memory Research — MICRO SRC 2023

SMASH: Co-designing Software Compression and Hardware-Accelerated Indexing for Efficient Sparse Matrix Operations [slides] — MICRO 2019

Academic Service

2027 — **HPCA 2027**, *Program Committee Member*

2026 — **MICRO 2026**, *Program Committee Member*

2026 — **4th Data Prefetching Championship (DPC-4)**, *Co-Organizer*
Held in conjunction with HPCA 2026, Sydney, Australia.

2026 — **HPCA 2026**, *Program Committee Member*
One of the first Ph.D. students invited to serve on the HPCA Program Committee.

2025 — **MICRO 2025**, *Artifact Evaluation Committee*

2025–2026 — **Virtuoso: Infrastructure for Evaluating Novel HW/OS Interfaces**, *Workshop & Tutorial Organizer*

- **MICRO 2025, Seoul** — First full-day edition, combining invited talks from industry and academia and guided implementation of address translation and memory-allocation HW/OS co-design techniques in Virtuoso [video]
- **ASPLOS 2026, Pittsburgh** — Second edition, featuring new talks and hands-on sessions with a focus on profiling OS routines and accelerating page-fault handling via HW/OS co-design techniques in Virtuoso [video]
- Third edition planned for MICRO 2026.

2019–2025 — **Sub-reviewer**
ISCA '19–'25 • MICRO '19–'25 • HPCA '19–'25 • ASPLOS '19–'25 • DSN '23 • DAC '21 • IEEE Micro Top Picks '19–'22 • EuroSys '23

Mentoring

- **Emre Hakan Demirli** (MSc, ETH Zurich) — Explored microarchitecture techniques to improve processor performance by representing loops in a symbolic fashion *[SyLo, Under Submission]*
- **Vlad-Petru Nitu** (MSc, ETH Zurich) — Exploring agent-based autonomous tools to discover performance overheads induced by OS routines *[Argus, Arch2.0 Workshop, ISCA 2026]*
- **Ipek Akdeniz** (MSc, ETH Zurich & TUM) — Exploring Hardware/OS co-design solutions to improve the reliability of robotics applications
- **Apostolis Kanellis** (MSc, NTUA) — Exploring memory-centric architectures for fast vision-language-action model inference
- **Georgios Gkikas** (MSc, NTUA) — Exploring GPU techniques for enabling real-time application execution for robotics and beyond
- **Harsh Songara** (MSc, ETH Zurich) — Designed flexible hardware substrates to avoid repeated changes in the DRAM/CPU command set and interface *[Terracotta, MICRO 2026]*
- **Spyros Galanopoulos** (MSc, NTUA) — Explored Hardware/OS cooperative solutions to accelerate memory allocation for short-lived workloads *[Valinor, arXiv 2026]*
- **Moritz Fechner** (MSc, ETH Zurich) — Explored alternative hash-based page table designs

- **Berkin Kerim Konar** (MSc, ETH Zurich) — Designed techniques to profile the overheads of memory allocation in emerging workloads
- **Ian Ganz** (BSc, ETH Zurich) — Explored the design of a new virtual memory simulation tool *[Virtuoso, ASPLOS 2025]*
- **Hong Chul Nam** (MSc, ETH Zurich) — Designed techniques to utilize underutilized processor resources to improve address translation *[Victima, MICRO 2023]*
- **Jeril Vathaloor** (MSc, ETH Zurich) — Designed an RL-based scheme that improves the effectiveness of swapping
- **Konstantinos Sgouras** (MSc, ETH Zurich) — Designed a new simulation infrastructure and methodology for evaluating HW/OS co-design techniques for memory management *[Virtuoso, ASPLOS 2025]*
- **Kosta Stojiljkovic** (MSc, ETH Zurich) — Explored new VM abstractions to reduce address translation overheads via hashing *[Utopia, MICRO 2023]*
- **Meryem Banu Cavlak** (BSc, Bilkent University) — Designed memory reordering techniques for sparse workloads

Teaching

2025–2026	**Selected Recorded Lectures: Virtual Memory & Prefetching**, *Lecturer* [Virtual Memory · Fall '25] [Prefetching II & VM · Spring '26] [VM Seminar · Spring '26] [Virtual Memory II · Spring '26]
2026–Present	**Computer Architecture (MSc), ETH Zurich**, *Instructor*
2020–2026	**Computer Architecture (MSc), ETH Zurich**, *Teaching Assistant*
2019–Present	**Design of Digital Circuits (BSc), ETH Zurich**, *Teaching Assistant*
2020–Present	**Seminar on Computer Architecture (BSc/MSc), ETH Zurich**, *Teaching Assistant*
2024–2025	**Fundamentals of Computer Architecture (BSc), ETH Zurich**, *Teaching Assistant*
2020–2023	**Project and Seminar on Intelligent Architectures, ETH Zurich**, *Organizer*
2018	**Advanced Computer Architecture, NTUA**, *Teaching Assistant*
2018	**Operating Systems, NTUA**, *Lab Assistant*

Languages

Greek Native | **French** C2 | **English** C2 | **German** A2